Multiple & Partial Correlation:-

Until now, how many variables were involved in the correlation coefficient?

Two Variables.

$$r_{12} = Corr(X_1, X_2)$$

Does this correlation consider the effect of other variables?

Ans. NO

It measures only the linear association between two variables.

Ex: Height and Weight

Suppose $r = 0.82$

Can we conclude that, Height alone explains weight?

Then say, This is why multivariate analysis introduces multiple and partial correlation.

Step:-2 Motivate with Three Variables:-

X_1 = Wright, X_2: Height, X_3: Age

Does age also effect weight?

If age influences both height and weight, is the simple correlation btw H & W, telling the whole story?

NO

This motivates partial correlation.

Step-3 Introduce Partial Correlation –

It means the relationship b/w two variables after removing the effect of one or more other variables.

$r_{12.3}$

Correlation b/w X_1 & X_2 given X_3.

We first remove the effect of age from both height & weight. Then correlate the remaining parts.

Eg. Suppose → Salary, Experience, Education,

Simple Correlation → Salary V/s Experience

But Education affects both –

Partial Correlation asks

What is the relationship b/w Salary & Experience after controlling education?

Introduce Multiple Correlation:-

Suppose, we want to predict weight using Height & age. Should we correlate weight seperately with Height?

NO We want both variables together.

Introduce $R_{1.23}$ → Multiple Correlation b/w X_1 & (X_2, X_3).

It measures how well a group of variables jointly predicts another variable.

Simple	Partial	Multiple Correlation
Bivariate Variables	Relationship b/w two variables after removing other variables.	one variable and a group of variables.

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

We are adjusting the original correlation by removing the influence of X_3.

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{13} r_{23}}{1 - r_{23}^2}}$$ This coefficient tells us how strongly X_1 is related to the

combination of X_2 and X_3.

Important Properties :-

Partial Correlation -

$-1 \leq r \leq 1$, May be positive or negative

If Zero → No linear association after controlling other variables.

Multiple Correlation :-

$0 \leq R \leq 1$, Never negative

Larger value means better joint prediction.

Q. Are Two Variables related? → Simple Correlation

Are Two Variables related after controlling another Variable? → Partial corr.

How ~~do~~ ~~with~~ well do several variables jointly explain one variable? → Multiple Correlation

Why Partial can be negative?

It's still a correlation between two variab after removing the effect of other variables.

For eg: $r_{12.3} = cor(X_1, X_2 | X_3)$

can be -0.7, 0, $+0.7$

Why? Because after controlling for X_3, the remaining parts of X_1 & X_2 can still move in either direction.

For eg: After removing the effect of age –

As the remaining part of X_1 ↑es, the remaining X_2 may decrease.

Therfore PC is negative.

Multiple can not be Negative:–

$$R_{1.23} = Corr(X_1, \overline{X_1})$$

$\overline{X_1}$ ⊥ (X_2 and X_3)

It is measuring how strongly X_1 is associated with its best linear prediction from the other variables.

$$R^2_{1.23} = \frac{\text{Explained var.}}{\text{Total var.}}$$

$$R_{1.23} = \sqrt{R^2_{1.23}} \qquad 0 \le R_{1.23} \le 1$$

It measures the strength of the joint linear relationship, not its direction.

Then Solve Numericals

* Multiple & Partial Correlation Coefficient:-

The Multiple correlation coefficient measures how well one dependent variable can be predicted using a set of independent variables.

Suppose X is a random vector with covariances matrix Σ.

$$X = \begin{bmatrix} \tilde{X}_{(1)} \\ \hline \tilde{X}_{(2)} \end{bmatrix} \begin{array}{l} \rightarrow \text{one variable among } p \text{ variable} \\ \rightarrow \text{other variables present} \end{array}$$

$$\Sigma_{p\times p} = \left[\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array}\right]$$

$$V(X_1) = \Sigma_{11}, \quad Cov(X_1, X_2) = \Sigma_{12}, \quad V(X_2) = \Sigma_{22}$$

So the Multiple correlation coefficient between $\tilde{X}_{(1)} = x_1$ (say) & $\tilde{X}_{(2)} = (x_2 \; x_3 \; \cdots \; x_p)'$ denoted by $R_{1.23\cdots p}$.

(indep | dependent)

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

(Indep | dependent)

The partial correlation coefficient between two variables (say X_1 & X_2) while controlling for other variables ($X_3, X_4, \cdots X_p$) measures the direct

other variables

$$r_{12.3} = \frac{r_{12} - r_{13}\, r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

Ques-① Let $X = [X_1 \;\; X_2 \;\; X_3]'$ be a r.v with correlation matrix $R = \left[\begin{array}{c|cc} 1 & 0.60 & 0.70 \\ \hline 0.60 & 1 & 0.65 \\ 0.70 & 0.65 & 1 \end{array}\right]$. Find

(i) Multiple correlation coefficient $R_{1.23}$

(ii) Partial correlation between X_1 & X_2. $r_{12.3}$

Sol: Here R matrix is given directly.

$$\therefore R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{13} r_{23}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{(0.6)^2 + (0.7)^2 - 2 \times (0.6)(0.7)(0.65)}{1 - (0.65)^2}}$$

$$= 0.725$$

(ii) $$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2}\sqrt{1 - r_{23}^2}} = \frac{0.60 - (0.70)(0.65)}{\sqrt{1 - (0.7)^2}\sqrt{1 - (0.65)^2}}$$

$$= 0.26714$$

Ques-② $R = \begin{bmatrix} 1 & 0.8 & 0.6 \\ 0.8 & 1 & 0.5 \\ 0.6 & 0.5 & 1 \end{bmatrix}$

(i) Find $R_{2.13}$ (ii) $r_{13.2}$.

$$R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2 r_{21} r_{23} r_{13}}{1 - r_{13}^2}}$$

Ques - From the following data, obtain $R_{1.23}$, $R_{2.13}$.

X_1	2	5	7	11
X_2	3	6	10	12
X_3	1	3	6	10

To obtain $R_{1.23}$ & $R_{2.13}$ we need r_{12}, r_{13}, r_{23}

For this we can use correlation formula as,

$$r_{12} = \frac{N(\Sigma X_1 X_2) - (\Sigma X_1)(\Sigma X_2)}{\sqrt{\{N(\Sigma X_1^2) - (\Sigma X_1)^2\}\{N(\Sigma X_2^2) - (\Sigma X_2)^2\}}}$$

r_{13}, r_{23} then put all values in formula.